Suggested solutions and discussion notes

Problem 1(a). Posterior. Multiplying the Poisson likelihood by the Gaussian prior gives

$$p(z \mid x) \propto \exp\left(xz - e^z - \frac{z^2}{2}\right).$$

The term $-e^z$ prevents the log posterior from being quadratic in z, so the posterior is not Gaussian.

Problem 1(b). ELBO. For a member $q = q_{\mu,v} \in \mathcal{Q}$, the expected log likelihood is

$$\mathbb{E}_q[\log p(x \mid z)] = x\mu - e^{\mu+v/2} - \log(x!).$$

Subtracting the prior KL gives

$$\boxed{\mathcal{L}(\mu, v; x) = x\mu - e^{\mu+v/2} - \log(x!) - \tfrac{1}{2}(\mu^2 + v - 1 - \log v).}$$

For the fixed model and observation,

$$\log p(x) = \mathcal{L}(q; x) + D_{\mathrm{KL}}[q(z) \| p(z \mid x)].$$

The evidence is constant with respect to μ, v, so maximising the explicit ELBO minimises reverse KL to the posterior without computing the evidence. The prior KL inside the objective is different from the posterior KL measuring approximation error.

Problem 1(c). Optimisation equations. Writing $b = e^{\mu+v/2}$, we obtain

$$\frac{\partial \mathcal{L}}{\partial \mu} = x - b - \mu, \qquad \frac{\partial \mathcal{L}}{\partial v} = \frac{1}{2}\left(\frac{1}{v} - 1 - b\right).$$

Thus an interior optimum satisfies

$$\boxed{\mu = x - e^{\mu+v/2}, \qquad v = \frac{1}{1 + e^{\mu+v/2}}.}$$

Use $v = e^{2a}$ with unconstrained log standard deviation a, or $v = e^{\ell}$ with unconstrained log variance ℓ. Either ensures $v > 0$. Students need not solve these nonlinear equations during the tutorial.

Tutor check. An independent numerical optimisation at $x = 4$ gives $\mu \approx 0.980645$ and $v \approx 0.248796$. Its ELBO is -3.075619; numerical integration gives log evidence -3.064399. Direct integration of posterior KL agrees with the gap 0.011221. These values are verification notes, not a student calculation.

Problem 2(a). A one-parameter ELBO. Under q_v, the sum has mean $2m$ and variance $2v$. Applying the identity $\mathbb{E}[(c-Y)^2] = (c-\mathbb{E}[Y])^2 + \mathrm{Var}(Y)$ gives

$$\mathbb{E}_{q_v}[(x - z_1 - z_2)^2] = (x - 2m)^2 + 2v.$$

The two scalar prior KLs add to $m^2 + v - 1 - \log v$. Hence

$$\begin{aligned}\mathcal{L}(v) &= -\frac{1}{2}\log(2\pi\tau^2) - \frac{(x-2m)^2 + 2v}{2\tau^2} - \left(m^2 + v - 1 - \log v\right) \\ &= C - \frac{v}{\tau^2} - v + \log v.\end{aligned}$$

Here m and τ are fixed. A single derivative gives

$$\mathcal{L}'(v) = -\tau^{-2} - 1 + v^{-1} = 0 \quad \Longrightarrow \quad \boxed{v^* = \frac{\tau^2}{1+\tau^2}.}$$

Since $\mathcal{L}''(v) = -1/v^2 < 0$, this is the unique maximum for $v > 0$.

Problem 2(b). Sum and difference. The measurement depends on S alone. Since D is independent of S and the measurement noise under the model, it is independent of x. Its posterior is still $\mathcal{N}(0, 2)$. In contrast, the scalar Gaussian update reduces the variance of S:

$$\mathrm{Var}_p(S \mid x) = \left(\frac{1}{2} + \frac{1}{\tau^2}\right)^{-1} = \frac{2\tau^2}{2+\tau^2} < 2, \qquad \mathrm{Var}_p(D \mid x) = 2.$$

Knowing the sum does not reveal how the two hidden quantities differ.

Problem 2(c). Analytic comparison. Independence and equal marginal variances under q_{v^*} imply

$$\mathrm{Var}_{q_{v^*}}(S) = \mathrm{Var}_{q_{v^*}}(D) = 2v^* = \frac{2\tau^2}{1+\tau^2}.$$

For every finite $\tau > 0$,

$$\boxed{\underbrace{\frac{2\tau^2}{2+\tau^2}}_{\mathrm{Var}_p(S|x)} < \underbrace{\frac{2\tau^2}{1+\tau^2}}_{\mathrm{Var}_q(S)=\mathrm{Var}_q(D)} < \underbrace{2}_{\mathrm{Var}_p(D|x)}.}$$

The first inequality compares equal positive numerators and different denominators. The second follows from $\tau^2 < 1 + \tau^2$. Thus q *overstates* uncertainty about the measured sum and *understates* uncertainty about the unmeasured difference.

As $\tau \to 0$, $\mathrm{Var}_q(D) \to 0$, although $\mathrm{Var}_p(D \mid x) = 2$ throughout. Correct posterior means do not guarantee correct uncertainty.

Discussion. Negative covariance lets one contribution increase while the other decreases, keeping the sum nearly fixed. With marginal variance u and covariance c, the sum and difference have variances $2(u+c)$ and $2(u-c)$; $c < 0$ allows a small sum variance and large difference variance. The exact posterior is Gaussian, so a full-covariance Gaussian family could represent it exactly.

Problem 3(a). Sampled ELBO. For scale $b = e^a$, $\mathbb{E}_q[z^2] = \mu^2 + 2b^2$ and the Laplace entropy is $1 + \log(2b)$. Subtracting this entropy from $-\mathbb{E}_q[\log p(z)]$ gives the supplied KL. Hence

$$\boxed{\widehat{\mathcal{L}}_K(\mu, a; x) = \frac{1}{K}\sum_{k=1}^{K} \ell\big(\mu + e^a \epsilon^{(k)}\big) - \left(\frac{\mu^2}{2} + e^{2a} - a + c_0\right).}$$

With $\epsilon^{(k)} \sim \text{Laplace}(0, 1)$, each transformed sample has distribution $q_{\mu,a} \in \mathcal{Q}$. At fixed parameters, the sample average has expectation $\mathbb{E}_q[\ell(z)]$ and the KL is exact, so the estimator is unbiased.

Problem 3(b). Implementation and fit. The three expressions to fill in are

Samples	`z = mu + np.exp(a) * eps`
Expected log likelihood estimate	`likelihood_term = np.mean(log_likelihood(z))`
Prior KL	`prior_kl = 0.5 * mu**2 + np.exp(2*a) - a + c0`

The notebook defines `c0` as $c_0 = \frac{1}{2}\log(2\pi) - 1 - \log 2$. Return `likelihood_term - prior_kl`. The supplied code checks the gradients and fits once with $K = 10$ and seed 0.

The prescribed run gives $\mu \approx 0.93894$ and scale $b \approx 0.40877$. Its numerically integrated ELBO is -2.425758, below log evidence -2.357307. Small differences across software versions are expected.

The Laplace fit has a pointed peak, exponential tails and symmetry. The reference is smooth, slightly asymmetric and has faster-decaying tails, so this family cannot match it exactly. A different family need not give a better approximation.

The family, sampling distribution and KL expression changed; the VI recipe is the same. We still maximise an ELBO over the family's parameters. The prior remains Gaussian even though q is Laplace.

Problem 3(c). Interpretation.

- **Updates need not improve the true ELBO.** Sampling can change the gradient's direction; an overly large step can also reduce the objective. The trace contains both parameter changes and sampling noise.
- **More samples do not change the family.** Increasing K reduces sampling variance at fixed parameters. The family $\mathcal{Q}$ still contains only Laplace distributions. More samples cannot remove the pointed peak, exponential tails or symmetry restriction.
- **A sampled estimate can exceed log evidence.** The actual ELBO obeys $\mathcal{L} = \log p(x) - D_{\text{KL}}[q \| p(z \mid x)] \leq \log p(x)$. For fixed q, an unbiased Monte Carlo estimate fluctuates around the true ELBO and may exceed $\log p(x)$. Unbiasedness is a statement about an expectation, not an inequality for every sample.